Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **object-oriented applications**, **JavaScript/TypeScript interfaces**, SQL-backed services, and **automated testing workflows** for dependable user-facing products.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**

- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.
- Achieved 99%+ reliability across 5K+ monthly jobs through **data validation and reconciliation**, error handling, cloud storage, CI/CD releases, and stable production monitoring for dependable user workflows reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ grading consistency across 10K+ submissions through **structured evaluation workflows**, persistent results, fallback paths, and instructor review controls supporting dependable education-facing production experiences reliably at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*

- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **automated testing and deployment**, implementing JWT access controls, WebSocket updates, Azure monitoring, Docker/Jenkins pipelines, and ARM templates across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**

- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.

Skills

Languages: Java, C++, Python, TypeScript, JavaScript, SQL, HTML, CSS
Engineering: Object-Oriented Programming, Data Structures, Algorithms, Design Patterns, REST APIs, Microservices, Git, CI/CD, Debugging, Agile
Web Applications: React, React Native, Node.js, FastAPI, Flask, WebSockets
Data: Azure SQL, PostgreSQL, MySQL, MongoDB, AWS RDS, Firebase, Data Modeling, Query Optimization
Testing & Delivery: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Jest, Selenium, Jenkins, Docker, Monitoring

Projects

Student Information Workflow - Design Concept

- Designed a **student-information workflow** concept with enrollment, attendance, grade, schedule, and family-access modules using object-oriented services, SQL relationships, role-based permissions, validation rules, audit history, and responsive interfaces.

School ERP Regression Suite - Design Concept

- Designed a **school ERP regression suite** concept covering payroll, purchasing, employee records, and approvals with reusable fixtures, automated integration tests, database assertions, defect reports, and release-gating workflows.

HMIS - Hospital Management Information System | Live

- Built a **TypeScript information platform** with consent-based record access, appointment workflows, Twilio alerts, and Sentry monitoring, reducing administrative workload 30% while supporting 100+ daily appointments reliably and securely.